

C-CLARKE INTERNATIONAL INSTITUTE OF DIGITAL SCIENCES

Source Code & Repository

Repository structure, branch workflow, and contribution history.

Program	Diploma in Advanced AI & Software Engineering
Batch	25/26
Project	Smart Hire — Service Marketplace
Team	Nilan Iddawela, Janaka Eranda, Dasun Anjana, Damith Lasantha, Matheesha Milan
Date	21 July 2026

1. Repository

Host	GitHub
URL	github.com/nilan92/Smart-Hire-System
Main branch	main
Active development branch	nilan (merged into main via reviewed pull requests)

2. Branch Structure

Development happened on a feature branch per contributor, merged into main through pull requests:

BRANCH	PURPOSE
main	Integrated, stable codebase
nilan	Backend APIs, admin, AI/MCP, messaging, QA, documentation
dev.janaka	Authentication, account status, database schema, dashboards
dev.dasun	Bookings, notifications, availability
feature/damith-ai / feature/damith-ai-backend	AI feature backend and recommendation logic
Matheesha	Landing page, services/categories/search backend
project_initialize	Initial project scaffold

3. Repository Structure

Smart-Hire-System/	
├─ backend/	FastAPI application
└─ app/	
└─ ai/	AI service, tools, prompts, chat router
└─ core/	config, database, security, dependencies
└─ models/	SQLAlchemy ORM models (16 tables)
└─ repositories/	data-access layer, one per domain
└─ routers/	FastAPI route definitions
└─ schemas/	Pydantic request/response models
└─ services/	business logic layer
└─ tests/	24 automated pytest tests
└─ mcp_server/	standalone MCP server
└─ alembic/	migration scaffold
├─ frontend/	Angular application
└─ src/app/	
└─ core/	services, models, guards, interceptors
└─ features/	customer / provider / admin pages
└─ shared/	shared components, layouts
├─ qa/	test plan, test cases, smoke & MCP tests
├─ documentation/	API contract, database design notes
├─ academic_documentation/	submission deliverables (this document set)
└─ postman/	Postman collection for manual API testing

4. Contribution Summary

Based on `git log --all` across every branch:

CONTRIBUTOR	COMMITTS	PRIMARY AREA
Nilan Iddawela	32	Backend APIs, admin, AI/MCP, messaging, QA, git workflow, documentation
Janaka Eranda	37	Authentication & account status, database schema, profile/dashboard pages
Dasun Anjana	28	Bookings, notifications, availability (frontend & backend)
Damith Lasantha	5	AI feature backend & recommendation logic
Matheesha Milan	4	Landing page, services/categories/search backend

5. Workflow

1. Each contributor worked on their own feature branch.
2. Changes were committed with descriptive messages explaining the “why”, not just the “what”.

3. Branches were merged into `main` through pull requests, reviewed before merging.
4. Merge conflicts (including a real one between two independently-updated documentation sets) were resolved deliberately rather than force-overwritten.